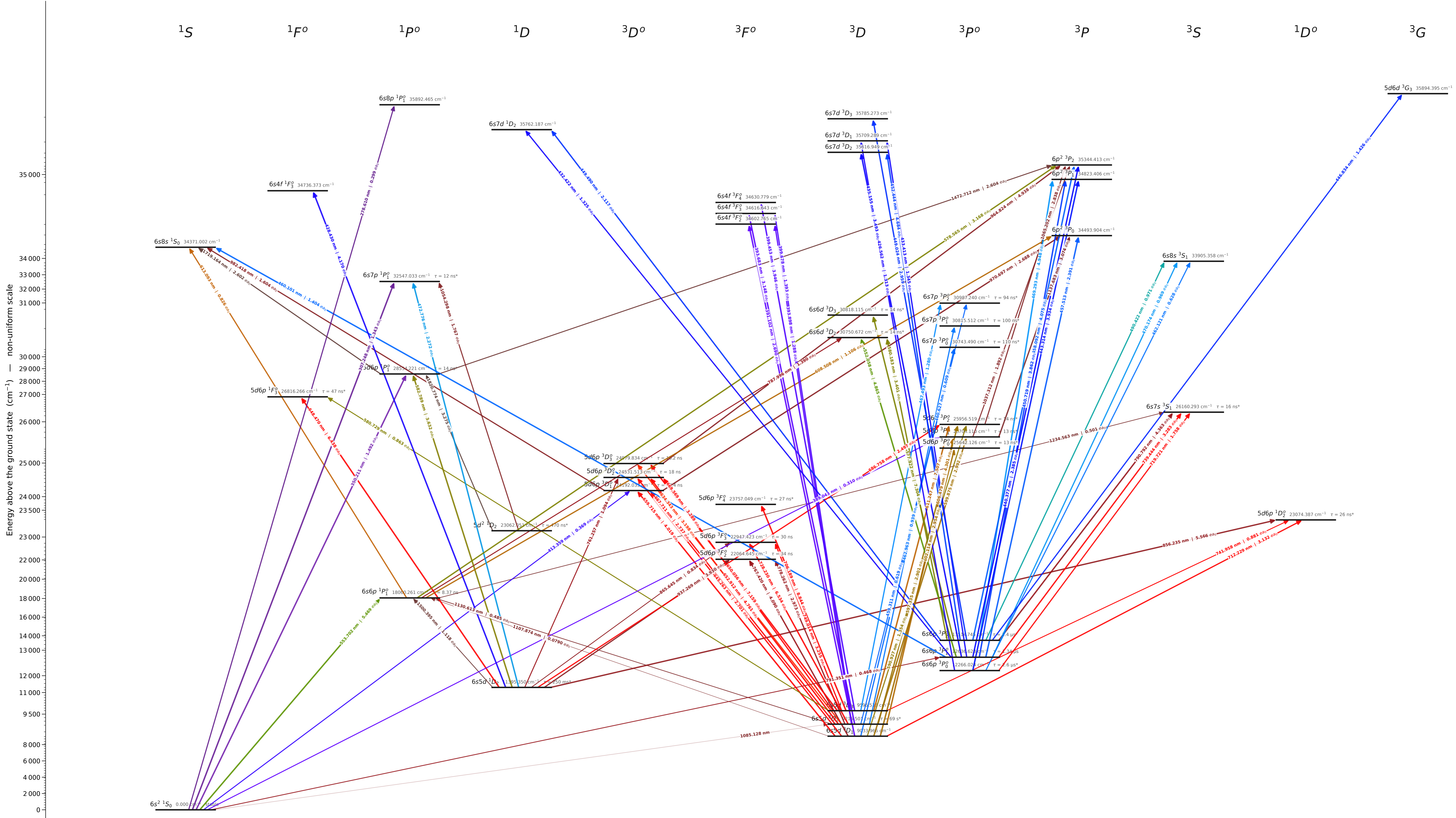
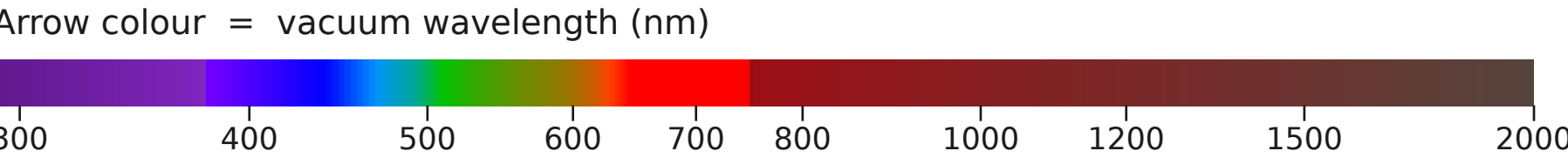


# Barium-137

<sup>137</sup>Ba (Ba I), nuclear spin  $I = 3/2$ . Energy levels and transitions below 2  $\mu\text{m}$ : 44 levels, 89 lines.

Ba <sup>+</sup> ionisation limit 42034.91 cm<sup>-1</sup> = 5.21166 eV (237.897 nm)

Arrow colour = vacuum wavelength | reduced dipole matrix element,  $|U||er||U|$  in ea<sub>0</sub>  
Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model or semi-empirical calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



## Key transitions

| Transition                                                                       | $\lambda$ vac (nm) | $\lambda$ air (nm) | $\nu$ (THz) | $ U  er  U $ (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | $\Gamma/2\pi$ up. (MHz) | branch (%) | Use                                                          |
|----------------------------------------------------------------------------------|--------------------|--------------------|-------------|---------------------------------|----------------------|-------------------------|------------|--------------------------------------------------------------|
| 6s <sup>2</sup> 1S <sub>0</sub> - 5d6p <sup>3</sup> D <sub>2</sub> <sup>o</sup>  | 413.3592           | 413.2427           | 725.2589    | 0.369                           | 1.300e+06            | 9.147                   | 2.3        | 413.36 nm intercombination line: fluorescence detection o... |
| 6s <sup>2</sup> 1S <sub>0</sub> - 6s6p <sup>1</sup> P <sub>1</sub> <sup>o</sup>  | 553.7019           | 553.5481           | 541.4330    | 5.469                           | 1.190e+08            | 19.01                   | 99.7       | main cooling / imaging line                                  |
| 6s5d <sup>3</sup> D <sub>1</sub> - 5d6p <sup>3</sup> D <sub>2</sub> <sup>o</sup> | 659.7147           | 659.5326           | 454.4274    | 4.019                           | 3.800e+07            | 9.147                   | 64         | 659.7 nm repumper                                            |
| 6s5d <sup>3</sup> D <sub>2</sub> - 5d6p <sup>3</sup> D <sub>2</sub> <sup>o</sup> | 667.7113           | 667.5270           | 448.9851    | 2.737                           | 1.700e+07            | 9.147                   | 32         | 667.7 nm repumper                                            |
| 6s <sup>2</sup> 1S <sub>0</sub> - 6s6p <sup>3</sup> P <sub>1</sub> <sup>o</sup>  | 791.3507           | 791.1330           | 378.8364    | 0.468                           | 2.990e+05            | 0.1183                  | 40.2       | intercombination line                                        |
| 6s <sup>2</sup> 1S <sub>0</sub> - 6s5d <sup>3</sup> D <sub>2</sub>               | 1085.1282          | 1084.8310          | 276.2738    | -                               | 1.454e-02            | 2.307e-09               | 100        | E2 decay limiting the 3D2 lifetime                           |
| 6s5d <sup>3</sup> D <sub>1</sub> - 6s6p <sup>1</sup> P <sub>2</sub> <sup>o</sup> | 1107.8743          | 1107.5709          | 270.6015    | 0.079                           | 3.100e+03            | 19.01                   | 0.0026     | 1107.8 nm decay branch / repumper                            |
| 6s5d <sup>3</sup> D <sub>2</sub> - 6s6p <sup>1</sup> P <sub>2</sub> <sup>o</sup> | 1130.6129          | 1130.3034          | 265.1592    | 0.485                           | 1.100e+05            | 19.01                   | 0.09       | 1130.6 nm decay branch / repumper                            |
| 6s5d <sup>1</sup> D <sub>2</sub> - 6s6p <sup>1</sup> P <sub>2</sub> <sup>o</sup> | 1500.3951          | 1499.9852          | 199.8090    | 1.118                           | 2.500e+05            | 19.01                   | 0.25       | 1500.4 nm decay branch of the cooling line / repumper        |

$\lambda$ ,  $\nu$  from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
 $\Gamma/2\pi$ : natural linewidth of the upper level,  $1/(2\pi\tau)$ .

## <sup>137</sup>Ba hyperfine structure ( $I = 3/2$ )

| Level                                         | A (MHz)     | B (MHz)    | Shift of each F level from the centre of gravity (MHz)         |
|-----------------------------------------------|-------------|------------|----------------------------------------------------------------|
| 6s5d <sup>3</sup> D <sub>1</sub>              | -520.536(3) | 17.890(3)  | F=1/2: +1323.70 F=3/2: +502.65 F=5/2: -776.33                  |
| 6s5d <sup>3</sup> D <sub>2</sub>              | 415.928(3)  | 25.899(13) | F=1/2: -1849.01 F=3/2: -1247.78 F=5/2: -224.15 F=7/2: +1254.26 |
| 6s5d <sup>3</sup> D <sub>3</sub>              | 456.559(4)  | 47.390(16) | F=3/2: -2710.92 F=5/2: -1605.07 F=7/2: -23.70 F=9/2: +2066.36  |
| 6s5d <sup>1</sup> D <sub>2</sub>              | -82.180(3)  | 59.564(14) | F=1/2: +421.93 F=3/2: +246.54 F=5/2: +3.86 F=7/2: -231.65      |
| 6s6p <sup>3</sup> P <sub>1</sub> <sup>o</sup> | 1150.590    | -41.610    | F=1/2: -2928.49 F=3/2: -1108.98 F=5/2: +1715.48                |
| 6s6p <sup>1</sup> P <sub>1</sub> <sup>o</sup> | -109.50(13) | 50.09(21)  | F=1/2: +336.36 F=3/2: +59.41 F=5/2: -151.73                    |
| 5d6p <sup>3</sup> D <sub>2</sub> <sup>o</sup> | 172.9(3)    | -3.5(6)    | F=1/2: -436.62 F=3/2: -169.40 F=5/2: +258.48                   |

Measured A, B constants; sources in the data file.

## Isotope shifts

| Transition                                                                       | $\lambda$ vac (nm) | Isotopes | Shift (MHz) |
|----------------------------------------------------------------------------------|--------------------|----------|-------------|
| 6s <sup>2</sup> 1S <sub>0</sub> - 5d6p <sup>3</sup> D <sub>2</sub> <sup>o</sup>  | 413.359            | 137-138  | 392.9(9)    |
| 6s <sup>2</sup> 1S <sub>0</sub> - 5d6p <sup>1</sup> D <sub>2</sub> <sup>o</sup>  | 413.359            | 136-138  | 178.1(8)    |
| 6s <sup>2</sup> 1S <sub>0</sub> - 5d6p <sup>3</sup> D <sub>2</sub> <sup>o</sup>  | 413.359            | 135-138  | 401.4(12)   |
| 6s <sup>2</sup> 1S <sub>0</sub> - 5d6p <sup>3</sup> D <sub>2</sub> <sup>o</sup>  | 413.359            | 134-138  | 124.3(13)   |
| 6s <sup>2</sup> 1S <sub>0</sub> - 6s6p <sup>1</sup> P <sub>1</sub> <sup>o</sup>  | 553.702            | 137-138  | 215.15(16)  |
| 6s <sup>2</sup> 1S <sub>0</sub> - 6s6p <sup>1</sup> P <sub>1</sub> <sup>o</sup>  | 553.702            | 136-138  | 128.0(4)    |
| 6s <sup>2</sup> 1S <sub>0</sub> - 6s6p <sup>1</sup> P <sub>2</sub> <sup>o</sup>  | 553.702            | 135-138  | 259.29(17)  |
| 6s <sup>2</sup> 1S <sub>0</sub> - 6s6p <sup>1</sup> P <sub>2</sub> <sup>o</sup>  | 553.702            | 134-138  | 143.0(5)    |
| 6s5d <sup>3</sup> D <sub>1</sub> - 5d6p <sup>3</sup> D <sub>2</sub> <sup>o</sup> | 659.715            | 137-138  | 189.2(10)   |
| 6s5d <sup>3</sup> D <sub>1</sub> - 5d6p <sup>3</sup> D <sub>2</sub> <sup>o</sup> | 659.715            | 136-138  | 112.9(10)   |
| 6s5d <sup>3</sup> D <sub>1</sub> - 5d6p <sup>3</sup> D <sub>2</sub> <sup>o</sup> | 659.715            | 135-138  | 228.3(10)   |
| 6s5d <sup>3</sup> D <sub>1</sub> - 5d6p <sup>3</sup> D <sub>2</sub> <sup>o</sup> | 659.715            | 134-138  | 125.0(20)   |
| 6s <sup>2</sup> 1S <sub>0</sub> - 6s6p <sup>3</sup> P <sub>2</sub> <sup>o</sup>  | 791.351            | 137-138  | 183.4(10)   |
| 6s <sup>2</sup> 1S <sub>0</sub> - 6s6p <sup>3</sup> P <sub>2</sub> <sup>o</sup>  | 791.351            | 136-138  | 109.2(10)   |
| 6s <sup>2</sup> 1S <sub>0</sub> - 6s6p <sup>3</sup> P <sub>2</sub> <sup>o</sup>  | 791.351            | 135-138  | 219.9(10)   |
| 6s <sup>2</sup> 1S <sub>0</sub> - 6s6p <sup>3</sup> P <sub>2</sub> <sup>o</sup>  | 791.351            | 134-138  | 122.3(25)   |
| 6s5d <sup>3</sup> D <sub>1</sub> - 6s6p <sup>1</sup> P <sub>1</sub> <sup>o</sup> | 1107.874           | 137-138  | -114(4)     |
| 6s5d <sup>3</sup> D <sub>1</sub> - 6s6p <sup>1</sup> P <sub>1</sub> <sup>o</sup> | 1107.874           | 136-138  | 59.3(6)     |
| 6s5d <sup>3</sup> D <sub>1</sub> - 6s6p <sup>1</sup> P <sub>1</sub> <sup>o</sup> | 1107.874           | 135-138  | -39(4)      |
| 6s5d <sup>3</sup> D <sub>1</sub> - 6s6p <sup>1</sup> P <sub>2</sub> <sup>o</sup> | 1107.874           | 134-138  | 144.1(10)   |
| 6s5d <sup>3</sup> D <sub>1</sub> - 6s6p <sup>1</sup> P <sub>2</sub> <sup>o</sup> | 1107.874           | 137-135  | -75.3(5)    |
| 6s5d <sup>3</sup> D <sub>2</sub> - 6s6p <sup>1</sup> P <sub>2</sub> <sup>o</sup> | 1130.613           | 137-138  | -69(3)      |

Shift =  $\nu$ (first isotope) -  $\nu$ (second isotope).